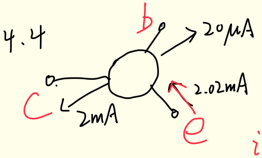
4.4

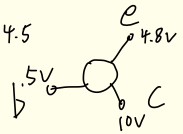


$i_E > i_C > i_B$

$\beta = \frac{i_c}{i_E} = \frac{2mA}{2\mu A} = 100.$

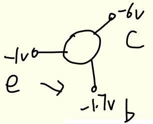
i_E流入，为NPN型

4.5



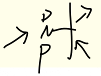
由5−4.8=0.2判定为锗管

C区电位最高，NPN型

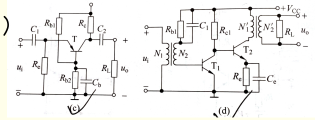


−1.7−(−1)=0.7 判为硅管.

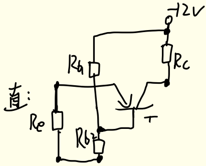
PNP型



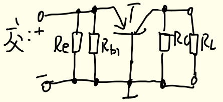
4.9. (c)



直：

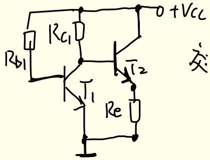


交：

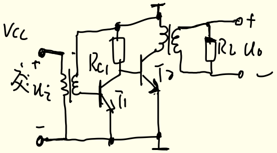


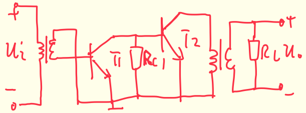
(D)

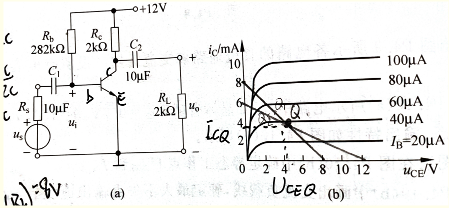
直：



交：







4.10

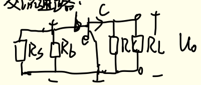
(1) 由KVL： $12 - R_c \cdot I_{CQ} = U_{CEQ}$ ① $\Rightarrow I_{BQ} \approx 40\mu A$.

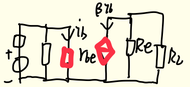
$12 - I_{BQ}R_b = U_{CEQ}$ ②

可得直流负载曲线，与$I_B = 40\mu A$交于Q为静态工作点

得$U_{CEQ} \approx 4V$. $I_{CQ} = 4mA$

(2) 交流通路：





$V_{CC}' = U_{CEQ} + I_{CQ}R_L'$

$= U_{CEQ} + I_{CQ}(R_c//R_L)$

$= 8V$

$U_{om} = \frac{4 - U_{BE}}{\sqrt{2}} \approx \underline{2.33V}$

(3)

$V_{CC}' - U_{CEQ} = 4V > U_{CEQ} - U_{BE} = \underline{3.3V}.$

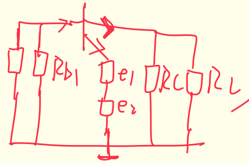
饱和

(4). $R_b\downarrow \rightarrow$ Q左移 $\rightarrow U_{om}\downarrow$

$R_c\uparrow \rightarrow$ Q左移. $\rightarrow U_{om}\downarrow$

$R_L\uparrow \rightarrow \begin{cases} Q \\ U_{om} \end{cases}$ 不变

直：



4.12

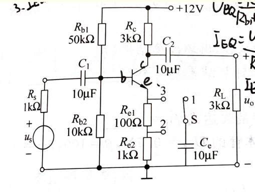


图 P4.12

$U_{BQ} = \frac{R_{b2}}{R_{b1}+R_{b2}} V_{CC} = 2V$

$I_{EQ} = \frac{U_{BQ} - U_{BE}}{R_{e1}+R_{e2}} \approx I_{CQ} \approx 1.182mA$

$I_{BQ} = \frac{I_{CQ}}{\beta} \approx 3.94\mu A$

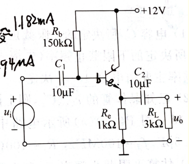


图 P4.13

4.11 (1) $I_{CQ} \approx 4mA$, $r_{bb'} + \beta\frac{U_T}{I_{CQ}} = (100 + 100\times\frac{26}{4})\Omega = 750\Omega$

$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{\beta(R_e//R_L)}{r_{be}} \approx -133.3$, $R_i = R_b//r_{be} \approx 750\Omega$

$\dot{A}_{us} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{U}_i}{\dot{U}_s}\cdot\frac{\dot{U}_o}{\dot{U}_i} = \frac{R_i}{R_i+R_s}\cdot\dot{A}_u \approx -57$, $\boxed{R_o = R_c = 2k\Omega}$

(2) $R_b\uparrow \rightarrow I_{BQ}\downarrow \rightarrow r_{be}\uparrow \rightarrow |\dot{A}_u|\downarrow \rightarrow R_i\uparrow$、$R_o$不变.

$R_s\uparrow \rightarrow \dot{A}_u$、$R_i$、$R_o$不变，$|\dot{A}_{us}|\uparrow$

$R_c\uparrow \rightarrow R_i$不变，$|\dot{A}_u|\uparrow$、$|\dot{A}_{us}|\uparrow$，$R_o\uparrow$

$R_L\uparrow \rightarrow R_i$、$R_o$不变，$|\dot{A}_u|\uparrow$、$|\dot{A}_{us}|\uparrow$

4.12.

(1) 设$V_{CC} = 12V$.

$U_{BQ} = \frac{R_{b2}}{R_{b1}+R_{b2}}\cdot V_{CC} = 2V$

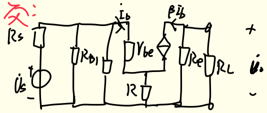
$I_{BQ} = \frac{I_{CQ}}{\beta} \approx 3.94\mu A.$

$I_{EQ} = \frac{U_{BQ} - U_{BE}}{R_{e1}+R_{e2}} \approx 1.182mA \approx I_{CQ}$

$U_{CEQ} = V_{CC} - I_{CQ}R_c - I_{EQ}(R_{e1}+R_{e2}) \approx 7.15V$

(2) $r_{be} = r_{bb'} + \beta\frac{U_T}{I_{CQ}} = (200 + 300\times\frac{26}{1.182}) = 6799\Omega$

交：



$\dot{A}_1 = \frac{\dot{U}_o}{\dot{U}_i} = -\frac{\beta(R_c//R_L)}{r_{be}+(1+\beta)(R_{e1}+R_{e2})}$ ①

$\dot{A}_2 = -\frac{\beta(R_c//R_L)}{r_{be}+(1+\beta)R_{e1}}$ ②

$\dot{A}_3 = \frac{\beta(R_c//R_L)}{r_{be}}$ ③

$\Rightarrow \dot{A}_1 \approx -1.33$

$\dot{A}_2 \approx -12.2$

$\dot{A}_3 \approx -66.2$

4.13 .

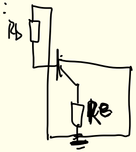
(1) $I_{BQ}=\dfrac{12-U_{BE}}{R_b+(1+\beta)R_e}$ ①

$I_{CQ}=\beta I_{BQ}$ ②

$U_{CEQ}=12-I_{CQ}R_e$ ③

$\Rightarrow\begin{cases}I_{BQ}\approx 28.2\mu A\\ I_{CQ}\approx 7.05mA\\ U_{CEQ}\approx 4.95V.\end{cases}$

直：



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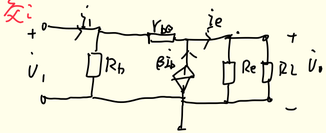

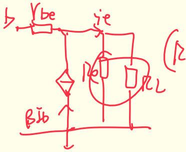

$(R_e//R_L)$ $(1+\beta)$

$r_{be}=r_{bb'}+\beta\dfrac{U_T}{I_{CQ}}\approx 1222\Omega$

$\dot{A}_u=\dfrac{\dot{U}_o}{\dot{U}_i}=\dfrac{(1+\beta)(R_e//R_L)}{r_{be}+(1+\beta)(R_e//R_L)}\approx 0.99.$

$R_i=R_b//[r_{be}+(1+\beta)(R_e//R_L)]\approx 83.7K\Omega$

$R_o=R_e//\dfrac{r_{be}}{1+\beta}\approx 4.86\Omega.$

[redacted]

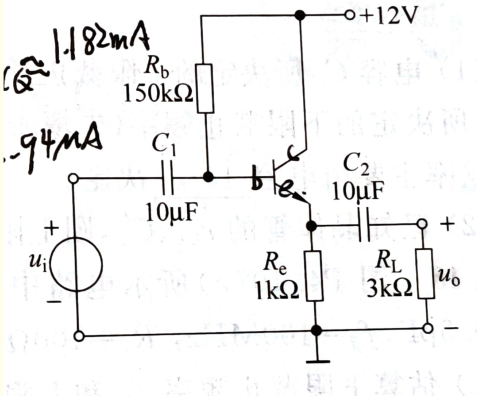

图 P4.13

电路等效分析，等效转化为对应Rbe电阻和电流源(受控)很重要，画好直流通路和交流通路，利用戴维宁等效能方便求出Ri，Ro，找到静态工作点，利于分析当某个元件变换时，其它元件的变化情况。